Do Liquidity Grabs and Fair Value Gaps Have a Statistical Edge?

Martingale · Research Note 004 *Author: Neil Gilani · Reproducible code:* `experiment.py`

Abstract

Two chart patterns central to modern online trading education — the **fair value gap (FVG)** and the **liquidity grab** — are tested on 2,903 days of SPY. Defining each mechanically and measuring the forward return after every occurrence, we find **no edge at the 1-day horizon** (every $p \geq 0.30$). At 5 days, three patterns look "significant," but the significance is an artifact: the returns are simply the market's upward drift, the supposedly *bearish* patterns are followed by the *largest positive* moves — the opposite of the claim — and the windows overlap. FVGs "fill" within 10 days 73–88% of the time, but bearish gaps fill *more*, which is exactly what drift does, not a magnet. Precisely defined and honestly tested, neither pattern predicts direction on daily SPY.

1. Introduction

A large share of online trading education is built on two ideas. A **fair value gap (FVG)** is a three-candle imbalance — a gap between the first candle's high and the third candle's low — said to act as a magnet that price returns to and respects. A **liquidity grab** (or stop hunt) is a move that pushes just past a recent high or low, triggering resting stop orders, before reversing — said to mark where "smart money" enters. Both are described with great confidence and tested almost never.

Hypothesis (stated before running it): consistent with Notes 001–003 and the efficient-markets baseline, neither pattern will show a meaningful, significant forward-return edge once defined mechanically. A clearly non-zero conditional return, significant after honest accounting, would refute this.

2. Data & Method

Data. Daily OHLC for SPY from January 2015 — 2,903 bars — via `yfinance`.

Definitions (mechanical, no discretion). - **Bullish FVG:** confirmed at candle 3 when `high[1] < low[3]`; bearish is the mirror (`low[1] > high[3]`). - **Liquidity grab, highs (bearish):** a new 20-day high printed intraday, but the day closes below the prior close. **Lows (bullish):** a new 20-day low swept, but the day closes up.

The edge test. Every event is confirmed at a day's close, and the forward return is measured from that close over 1 and 5 trading days — no lookahead. We report the mean forward return, a t-statistic and two-sided p-value versus zero, and the hit rate, against the market's unconditional baseline over the same horizon. The 1-day horizon is primary (consecutive windows barely overlap); the 5-day is secondary, with overlapping windows that inflate significance.

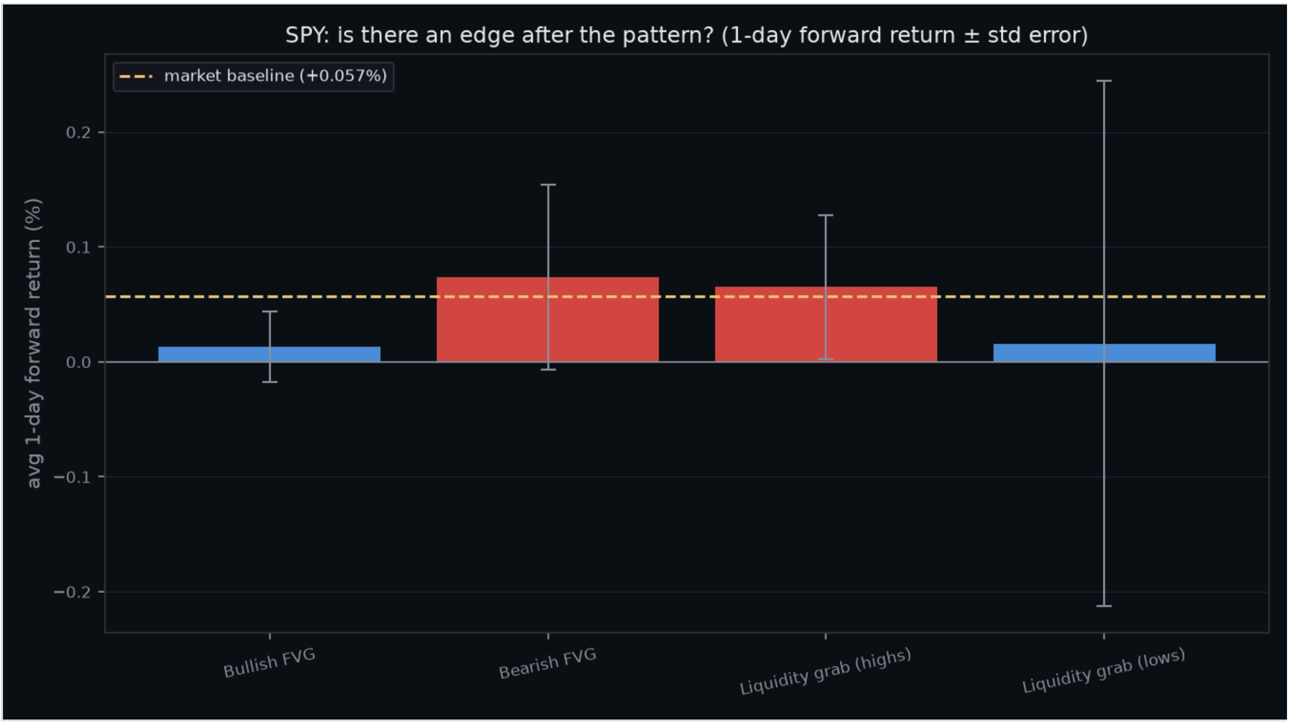
3. Results

Events found across 2,903 bars: 687 bullish FVGs, 362 bearish FVGs, 159 liquidity-grab-highs, 49 liquidity-grab-lows.

1-day forward return — baseline +0.057%:

Pattern	n	Mean fwd	*t*	*p*	Hit rate
Bullish FVG	687	+0.013%	+0.42	0.672	52%
Bearish FVG	362	+0.074%	+0.92	0.357	53%
Liquidity grab (highs)	159	+0.065%	+1.04	0.298	56%
Liquidity grab (lows)	49	+0.016%	+0.07	0.944	57%

Nothing is significant, and nothing clears the baseline by a meaningful margin.



5-day forward return — baseline +0.284%:

Pattern	n	Mean fwd	*t*	*p*	Hit rate
Bullish FVG	687	+0.165%	+2.43	0.015	61%
Bearish FVG	361	+0.384%	+2.46	0.014	62%

Pattern	n	Mean fwd	t	p	Hit rate
Liquidity grab (highs)	159	+0.391%	+3.43	0.001	63%
Liquidity grab (lows)	49	+0.305%	+0.71	0.476	47%

FVG fill rate within 10 days: bullish 73%, bearish 88%.

4. Discussion

At the clean 1-day horizon there is nothing to discuss: no pattern is significant, and none beats the market's ordinary drift. The interesting part is the 5-day column, because three of its p-values fall below 0.05 — and every one of them is a trap worth naming.

The "significant" numbers are just the market's drift. The 5-day tests are against zero, but SPY drifts up +0.284% over an average 5 days. Bullish FVG returns +0.165% — *below* the baseline. A signal that is "significantly positive" yet underperforms doing nothing is not an edge; it is the drift, weakened.

The bearish patterns point the wrong way. Bearish FVGs and liquidity-grab-highs are supposed to precede *down* moves. They post the *largest positive* returns on the board (+0.384%, +0.391%). If anything they slightly *out*-drift the market — the exact opposite of the claim. A pattern that predicts the reverse of what it promises is not a weak signal; it is no signal, riding the same upward tide as everything else.

Overlap and multiple testing finish the job. The 5-day windows overlap across nearby events, which inflates the t-statistics (nearby observations are not independent). And we ran eight tests; at $p < 0.05$ roughly 0.4 false positives are expected by chance, more once the tests are correlated. A few readings near $p \approx 0.01$ across overlapping, drift-dominated tests is exactly what noise looks like once you account for how you looked.

The fill rate confirms nothing. FVGs fill 73–88% of the time within ten days, which sounds like vindication until you notice the direction of the asymmetry: *bearish* gaps fill more often (88%) than bullish ones (73%). That is precisely what an upward-drifting market does — it rises and fills the gaps left below it — and has nothing to do with a magnet. On a near-random walk price revisits nearby levels constantly; without a matched null, a high fill rate is not evidence of a force, and the asymmetry actively points to drift instead.

Put together: every apparent signal dissolves into market drift, a wrong-direction result, or an overlap/multiple-testing artifact. After even the 1 bp costs of Note 003, nothing here is tradeable.

5. Limitations

This tests one instrument (SPY) on **daily** bars, whereas these patterns are most often claimed on intraday timeframes and on futures or FX; a daily null result falsifies the daily-SPY version, not every possible version. The definitions are one reasonable mechanisation of concepts practitioners apply with discretion and context (trend, session, confluence) — though that discretion is also where hindsight and selection quietly enter. The 5-day results use overlapping windows, which overstate significance. And correlation is not causation.

6. Conclusion

We defined fair value gaps and liquidity grabs mechanically and asked whether they predict the next move on daily SPY. At one day, they do not — no pattern is significant. At five days, the only "significant" results are the market's own drift, are largest for the patterns pointing the wrong way, and are inflated by overlapping windows and multiple testing. The gaps "fill" often simply because price wanders back regardless. These patterns are real as descriptions of what price *did*; the data gives no reason to believe they predict what it will do.

References

- Fama, E. (1970). *Efficient Capital Markets.*
- Harvey, C., Liu, Y. & Zhu, H. (2016). …*and the Cross-Section of Expected Returns.* (On how many "discovered" signals fail to replicate.)
- Martingale, Research Notes 001–003.

How to cite

> Gilani, N. (2026). *Do Liquidity Grabs and Fair Value Gaps Have a Statistical Edge?* Martingale, Research Note 004. https://github.com/hilothefunnydog123-coder/quant-research